

THUSHAN MALAKA RANASINGHE

[!\[\]\(c8d96c8885d3000a912c2582004aed63_img.jpg\) LinkedIn](#) | [!\[\]\(3ad821e3ca7dd4cb7003e9c8d982e254_img.jpg\) Gampaha, Sri Lanka](#) | [!\[\]\(177bde115c7ebbeffa559d05eea9e94b_img.jpg\) +94 717886186](#) | [!\[\]\(cab2e95699b614c49dd80341e1932607_img.jpg\) Thushan](#) | [!\[\]\(75f9a43febaa9aa08b77b73c8ad8a855_img.jpg\) thushanmalaka123@gmail.com](#) | [!\[\]\(cc11c320e6649662ebbe761994792de4_img.jpg\) GitHub](#)

Profile

Results-driven Computer Engineering graduate with a strong foundation in building production grade web applications and enterprise solutions. Proven expertise in architecting real time systems and migrating legacy workflows to modern tech stacks like Go, NestJS, React and NextJS. Published researcher in Cloud Workload Prediction with a deep interest in optimizing system reliability and automation through CI/CD and Docker.

Skills

- **Programming Languages:** TypeScript, JavaScript, Python, Java, C++, Go
- **Frontend Development:** React.js, Next.js, React Native, TailwindCSS, Zustand, Vite, Redux
- **Backend Development:** Node.js, NestJS, Express.js, Socket.io
- **Databases and Cloud:** PostgreSQL, MongoDB, MySQL, SQLite
- **DevOps & Automation:** Docker, Docker Hub, GitHub Actions, CI/CD Pipeline
- **Tools & Methodologies:** Git, GitHub, Jira, Agile, Postman
- **Specialized Knowledge:** Machine Learning, AI

Work Experience

- **Software Engineering Intern - Sri Lanka Telecom (SLT) Colombo, Sri Lanka**
Mar 2025 – Sep 2025
 - sDesk - Led the migration of a PHP-based workflow to a React/Redux and NestJS architecture, implementing role-based access control and comprehensive features lacking in the previous system. Eliminated manual delays by integrating Socket.io for instantaneous ticket assignments and updates, while securing enterprise access via Azure AD Single Sign-On (SSO).
- **Freelancing**
Oct 2025 – Dec 2025
 - HR Management system – Full stack system using React and Nodejs to centralize employee records and automate payroll calculation and leave management to reduce administrative overhead.
- **Software Engineering Intern – Lead Innovationz Horana, Sri Lanka**
Dec 2025 – Mar 2026
 - myParcel - Reduced operational costs associated with Google Maps API by migrating the mapping infrastructure to OpenStreetMap with a Google Tiles interface.
 - LI Board System - Implemented a robust data protection strategy for databases by implementing Cron-based automated and manual backups feature for Li Board System.
 - Gamage Latex – Developed a full stack rubber management system using Go and React.

Education

- **Bachelor of the Science of Engineering (BSc Eng) in Computer Engineering**
University of Jaffna (Apr 2021 – Nov 2025)

Publications

- **Workload Prediction in Cloud Environment: Study of Temporal Convolutional Networks and Hybrid Deep Learning Models**

- 9th IEEE International Research Conference on Smart Computing and Systems Engineering (SCSE) — 2026.
- Presented a comparative analysis of TCN and Hybrid Deep Learning models for optimizing cloud resource management.
- Paper accepted and presented and currently in the process of publication in IEEE Xplore.

Projects

- **Gamage Latex(ERP System)** :

- Engineered a full-stack Go/Gin and React ERP system to automate latex intake, billing and monthly settlements, replacing manual record-keeping with a transactional processing engine.
- Implemented secure S3-compatible storage (Cloudflare R2) using pre-signed URLs for direct asset uploads and integrated JWT for authentication.

Key Technologies : Go (Gin), PostgreSQL, GORM, Cloudflare R2, TypeScript, React, Docker

- **TMchat (Link)** :

- Developed a full-stack MERN application (Vite/React) featuring instant messaging via Socket.io, managed state with Zustand, and integrated Cloudinary for dynamic profile management.
- Implemented a secure JWT/HTTP-only cookie authentication system, incorporating Arcjet for bot protection and rate limiting, while ensuring responsive design with TailwindCSS.

Key Technologies : Node, Express, MongoDB, Socket.io, JWT, Cloudinary, Arcjet, Nodemailer, React, Vite, TailwindCSS, DaisyUI

- **TMdent (Link)**:

- Developed a modern full-stack dental appointment platform using Next.js, PostgreSQL, and Prisma for type-safe database management.
- Integrated Clerk for secure authentication and implemented automated workflows using Nodemailer for email confirmations and Vapi for AI-driven voice interactions.

Key Technologies : Next.js, React, PostgreSQL, Prisma, Clerk, TanStack Query, TailwindCSS, Nodemailer, Vapi

- **Ameliorate [Open Source Contribute] (Link)** :

- Developed a TypeScript/Prisma cleanup script using raw SQL to identify and purge orphaned records, ensuring relational data consistency across complex graph nodes and edges..
- Developed a GitHub Action for automated weekly database cleanup with a manual override capability to ensure data integrity.

Key Technologies : TypeScript, Prisma, SQL, GitHub Actions (CI/CD), Node.js

- **Beat Music (Link)** :

- Created a music platform using Next.js, TypeScript and TailwindCSS.
- Implemented secure client-side session management via Firebase Auth and developed Next.js API routes to proxy search and song metadata from JioSaavn and YouTube.

Key Technologies : Next.js, TypeScript, TailwindCSS, Firebase Authentication

References

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